

CHAPTER 6.3

VOLUMES BY CYLINDRICAL SHELLS

CYLINDRICAL SHELLS

6.3.1 Volume Problem

Let f be continuous and nonnegative on $[a, b]$ for $0 \leq a < b$, and let R be the region that is bounded above by $y = f(x)$, below by the x -axis, and on the sides by the lines $x = a$ and $x = b$. Find the volume V of the solid of revolution S that is generated by revolving the region R about the y -axis (Figure 6.3.1).

Sometimes problems of the above type can be solved by the method of disks or washers perpendicular to the y -axis, but when that method is not applicable or the resulting integral is difficult, the method of cylindrical shells, which we will discuss here, will often work. A **cylindrical shell** is a solid enclosed by two concentric right circular cylinders (Figure 6.3.2).

6.3.2 Volume by Cylindrical Shells About the y -axis

Let f be continuous and nonnegative on $[a, b]$ for $0 \leq a < b$, and let R be the region that is bounded above by $y = f(x)$, below by the x -axis, and on the sides by the lines $x = a$ and $x = b$. Then the volume V of the solid of revolution that is generated by revolving the region R about the y -axis is given by

$$V = \int_a^b 2\pi x f(x) dx \quad (1)$$

Example 1

Use cylindrical shells to find the volume of the solid generated when the region enclosed between $y = \sqrt{x}$, $x = 1$, $x = 4$, and the x -axis is revolved about the y -axis.

Solution: See Textbook page 434.

VARIATIONS OF THE METHOD OF CYLINDRICAL SHELLS

The method of cylindrical shells is applicable in a variety of situations that do not fit the conditions required by Formula (1). For example, the region may be enclosed between two curves, or the axis of revolution may be some line other than the y -axis. However, rather than develop a separate formula for every possible situation, we will give a general way of thinking about the method of cylindrical shells that can be adapted to each new situation as it arises.

6.3.3 An Informal Viewpoint about Cylindrical Shells

The volume V of a solid of revolution that is generated by revolving a region R about an axis can be obtained by integrating the area of the surface generated by an arbitrary cross section of R taken parallel to the axis of revolution.

The following examples illustrate how to apply this result in situations where Formula (1) is not applicable.

Example 2

Use cylindrical shells to find the volume of the solid generated when the region R in the first quadrant enclosed between $y = x$ and $y = x^2$ is revolved about the y -axis ([Figure 6.3.8a](#)).

Solution: See Textbook page 435.

Example 3

Use cylindrical shells to find the volume of the solid generated when the region R under $y = x^2$ over the interval $[0, 2]$ is revolved about the line $y = -1$.

Solution: See Textbook page 435.

EXERCISE SET 6.3

Use cylindrical shells to find the volume of the solid generated when the region enclosed by the given curves is revolved about the y -axis.

5. $y = x^3$, $x = 1$, $y = 0$

Solution: The volume of the solid is

$$V = \int_a^b 2\pi x f(x) dx = 2\pi \int_0^1 x^4 dx = 2\pi \left[\frac{x^5}{5} \right]_0^1 = \frac{2\pi}{5}$$

7. $y = \frac{1}{x}$, $y = 0$, $x = 1$, $x = 3$

Solution: The volume of the solid is

$$V = \int_a^b 2\pi x f(x) dx = 2\pi \int_1^3 x(1/x) dx = 2\pi \left[x \right]_1^3 = 4\pi$$

9. $y = 2x - 1$, $y = -2x + 3$, $x = 2$

Solution: The volume of the solid is

$$\begin{aligned} V &= \int_a^b 2\pi x f(x) dx = 2\pi \int_1^2 x[(2x - 1) - (-2x + 3)] dx \\ &= 2\pi \int_1^2 x(4x - 4) dx = \frac{20\pi}{3} \end{aligned}$$

11. $y = \frac{1}{x^2 + 1}$, $x = 0$, $x = 1$, $y = 0$

Solution: The volume of the solid is

$$\begin{aligned} V &= \int_a^b 2\pi x f(x) dx = 2\pi \int_0^1 x \times \frac{1}{x^2 + 1} dx \\ &= 2\pi \int_0^1 \frac{x}{x^2 + 1} dx = \pi \left[\ln(x^2 + 1) \right]_0^1 = \pi(\ln 2) \end{aligned}$$

13. $y^2 = x$, $y = 1$, $x = 0$

Solution: The volume of the solid is

$$\begin{aligned} V &= \int_c^d 2\pi y f(y) dy = 2\pi \int_0^1 y(y^2) dy \\ &= 2\pi \left[\frac{y^4}{4} \right]_0^1 = \frac{\pi}{2} \end{aligned}$$

15. $y = x^2$, $x = 1$, $y = 0$

Solution: The volume of the solid is

$$\begin{aligned} V &= \int_c^d 2\pi y f(y) dy = 2\pi \int_0^1 y(1 - \sqrt{y}) dy \\ &= 2\pi \left[\frac{y^2}{2} - \frac{2}{5} y^{5/2} \right]_0^1 = 2\pi \left(\frac{1}{2} - \frac{2}{5} \right) = \frac{\pi}{5} \end{aligned}$$

29. Use cylindrical shells to find the volume of the solid that is generated when the region that is enclosed by $y = \frac{1}{x^3}$, $x = 1$, $x = 2$, $y = 0$ is revolved about the line $x = -1$.

Solution: The volume of the solid is

$$\begin{aligned} V &= \int_c^d 2\pi x f(x) dx = 2\pi \int_1^2 (x+1) \left(\frac{1}{x^3} \right) dx \\ &= 2\pi \int_1^2 \left(\frac{1}{x^2} + \frac{1}{x^3} \right) dx \\ &= 2\pi \left[-\frac{1}{x} - \frac{1}{2x^2} \right]_1^2 = \frac{7\pi}{4} \end{aligned}$$

Note: For Figures, please follow my class lectures.